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Highly accurate simplified lattice Boltzmann method

Z. Chen, C. Shu,[a] and D. Tan
Department of Mechanical Engineering, National University of Singapore, 10 Kent Ridge Crescent, Singapore 119260, Singapore

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In this paper, we present a highly accurate simplified lattice Boltzmann method (HSLBM) which can achieve the third-order of accuracy in space. By introducing virtual streaming nodes and decoupling the streaming distance from the mesh spacing, HSLBM effectively combines the local second-order simplified and highly stable lattice Boltzmann method (SHSLBM) and the overall high order scheme. The correlation between the streaming distance and the mesh spacing is accomplished by a high-order Lagrange interpolation algorithm. Through a series of tests, it is found that using 5 interpolation points and setting streaming distance as 1/5 of the mesh spacing can give optimal results. In general, HSLBM improves the accuracy of SHSLBM from the second order to the third order, while maintaining its merits like low memory cost, convenient implementation of physical boundary conditions, and good numerical stability. To match the accuracy of the flow solver, a linear extrapolation scheme with the third-order of accuracy is also proposed to determine the boundary values of the non-equilibrium distribution functions. Various benchmark tests are performed to demonstrate the robustness of HSLBM in simulating two- and three-dimensional incompressible viscous flows as well as its flexibility in problems with curved boundaries and on body-fitted meshes. *Published by AIP Publishing.* https://doi.org/10.1063/1.5050185

I. INTRODUCTION

The Lattice Boltzmann Method (LBM) is a popular numerical method in the mesoscopic scale.[1–15] The evolution of LBM is reflected by the streaming-collision process which only involves simple and explicit algebraic equations. The simplicity and explicitness earn LBM a great popularity in the academic community and for engineering applications. However, various drawbacks of LBM have been discovered in its application, including high cost in virtual memory, inconvenient implementation of physical boundary conditions, and poor numerical stability at high Reynolds number.[16–19]

Various endeavors have been made to eliminate the above drawbacks of conventional LBM, especially to improve its numerical stability.[20] The first solution was proposed by using multiple-relaxation-time (MRT) tuned with the linear stability analysis.[4] The entropic LBM with the Bhatnagar-Gross-Krook (BGK) collision model method appears as the second solution and was conceived based on the maximization of the entropy, by locally tuning the BGK single relaxation time.[21–23] The third alternative is to work with regularized LB equations.[24–26] In this approach, the LB equation is rewritten in a manner as to filter the undesirable ghost moments from the numerical scheme, which could effectively suppress numerical instability in the conventional LBGK model.

A recent exploration within the LB framework is the simplified and highly stable lattice Boltzmann method (SHSLBM).[27–30] SHSLBM is a second-order scheme developed by reconstructing solutions to macroscopic equations recovered from the lattice Boltzmann equation (LBE) and resolved in a predictor-corrector scheme. The resultant formulations of SHSLBM are still simple and explicit algebraic equations, and show direct updating of macroscopic variables rather than the distribution functions. Such a characteristic favors SHSLBM in reduction of virtual memory cost and implementation of physical boundary conditions. Moreover, through von Neumann stability analysis, SHSLBM can be theoretically proven to be much more stable than the conventional LBM. Despite the above merits, one remaining constraint of SHSLBM is its second-order accuracy. Although SHSLBM shows nice numerical stability and can achieve converged results at coarse meshes, the obtained results are low in accuracy due to the second-order of spatial accuracy. To solve this issue, we consider adopting more accurate scheme within the SHSLBM framework.

The numerical error, defined as $O(h^k)$, is affected by two factors.[31] One is the mesh spacing h. And the other is the order of accuracy k. These two factors correspond to two approaches of reducing numerical error. The first approach is to refine the mesh size. This attempt is simple in implementation but inevitably lowers computational efficiency. Alternatively, adopting schemes with a higher order of accuracy can give lower numerical error for the same mesh size. However, this strategy is usually accompanied by more complicated formulations and lower flexibility in dealing with non-uniform meshes and curved boundaries. In the present work, we try to explore a novel highly accurate scheme which could effectively combine advantages of the above two approaches in reducing numerical error.

Developing a high-order-of-accuracy scheme within the conventional LBM framework is a very challenging task which requires improvement of accuracy in the discretization of the velocity space, the streaming term, and the spatial

[a]Author to whom correspondence should be addressed: mpeshuc@nus.edu.sg.

derivatives.[32,33] Such a strategy, if possible, is accompanied by more expansion terms which could sacrifice the explicitness and simplicity of the method. Existing "high-order" schemes in the LBM framework generally mean recovering high-order moments rather than achieving high-orders of spatial accuracy.[7,34,35]

To increase the accuracy while maintaining the simplicity, we propose a highly accurate simplified lattice Boltzmann method (HSLBM) in this paper. The basic idea of HSLBM is to distribute virtual streaming nodes to decouple the streaming distance δ_x from the mesh spacing h; i.e., δ_x is not equal to h. Using regular formulations of SHSLBM, the local streaming accuracy is maintained in the order of $O({\delta_x}^2)$; i.e., the numerical error is $E_1 = a{\delta_x}^2$. Note that the overall accuracy is assessed by $O(h^k)$, i.e., $E_2 = bh^k$. Here a and b are two parameters in the same order of $O(1)$. Since E_1 and E_2 are two expressions of the same evolution error, they should be equal in quantity. Therefore, by reducing the value of δ_x, it is possible to make k larger than 2, which means that the higher order of overall accuracy can be obtained by locally using regular formulations of SHSLBM with small δ_x. Such treatment brings another issue, which is the correlation between the streaming distance and the mesh spacing. Since the streaming nodes are no longer located on mesh points, an interpolation algorithm is required to determine physical properties on the streaming nodes. Therefore, the overall accuracy of HSLBM is determined by both the setting of streaming distance and the order of accuracy in interpolation. In order not to sacrifice the overall accuracy, a high-order interpolation scheme should be applied. We will show in this paper that when Lagrange interpolation is adopted, using 5 interpolation points and setting $\delta_x = h/5$ give optimal results, and an overall third-order of accuracy in space can be achieved. In general, HSLBM can be understood as an effective combination of locally second-order reconstruction scheme and overall high order scheme that is correlated with a high-order interpolation algorithm.

Various merits can be expected in HSLBM. From a physical perspective, the present HSLBM is still developed within the LBM framework, which ensures its physical robustness. From a numerical perspective, HSLBM increases the overall accuracy of SHSLBM while inheriting its simplicity, explicitness, and good numerical stability. Like SHSLBM, HSLBM also directly tracks the evolution of macroscopic variables, which lowers the cost in virtual memory and facilitates the implementation of physical boundary conditions.

The remaining parts of the paper are laid out as follows: Sec. II briefly recalls the formulations of the conventional LBM and recently developed SHSLBM scheme. Section III discusses the formulations, interpolation algorithm, and boundary treatments in HSLBM. Detailed discussion on the settings of parameters in HSLBM is presented in Sec. IV. Section V gives five representative numerical examples, including plane Poiseuille flow, 2D lid-driven cavity flow, polar cavity flow, flow past a stationary circular cylinder, and 3D lid-driven cavity flow, for comprehensive demonstration of the robustness of HSLBM as well as its flexibility in problems with curved boundaries and on body-fitted meshes. Conclusions are drawn in Sec. VI.

II. SIMPLIFIED AND HIGHLY STABLE LATTICE BOLTZMANN METHOD (SHSLBM)

A. From lattice Boltzmann equation (LBE) to Navier-Stokes (N-S) equations

The lattice Boltzmann equation with BGK approximation (LBGK model)[3] can be expressed as

$$f_\alpha(\mathbf{r}+\mathbf{e}_\alpha\delta_t, t+\delta_t) = f_\alpha(\mathbf{r},t) + \frac{f_\alpha^{eq}(\mathbf{r},t) - f_\alpha(\mathbf{r},t)}{\tau}, \quad \alpha = 0,1,\ldots,N, \tag{1}$$

where N and τ are the number of the streaming directions and the single relaxation parameter, respectively. δ_t denotes the streaming time step, which is set equal to the streaming distance δ_x in most cases. f_α denotes the density distribution function along the streaming direction of α and ${f_\alpha}^{eq}$ is the equilibrium distribution function defined as

$$f_\alpha^{eq} = \rho\omega_\alpha\left[1 + \frac{\mathbf{e}_\alpha\cdot\mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_\alpha\cdot\mathbf{u})^2 - (c_s|\mathbf{u}|)^2}{2c_s^4}\right], \tag{2}$$

where ρ, $\mathbf{u}$, and c_s denote the density, the velocity vector, and the sound speed, respectively; and $\mathbf{e}_\alpha$ is the particle velocity in the α direction. Distribution functions are related to the macroscopic variables through conservation laws, i.e.,

$$\rho = \sum_{\alpha=0}^{N} f_\alpha = \sum_{\alpha=0}^{N} f_\alpha^{eq}, \quad \rho\mathbf{u} = \sum_{\alpha=0}^{N} f_\alpha\mathbf{e}_\alpha = \sum_{\alpha=0}^{N} f_\alpha^{eq}\mathbf{e}_\alpha. \tag{3}$$

Through Chapman-Enskog (C-E) expansion analysis, the lattice Boltzmann equation (1) can be proven to recover the following macroscopic equations with the second order accuracy in space and time:[2,36]

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\left(\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\right) = 0, \tag{4}$$

$$\frac{\partial\rho\mathbf{u}}{\partial t} + \nabla\cdot\left[\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{eq} + \left(1-\frac{1}{2\tau}\right) \times \sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}\right] = 0, \tag{5}$$

where $(\mathbf{e}_\alpha)_\beta$ and $(\mathbf{e}_\alpha)_\gamma$ are the components of the lattice velocity vector $\mathbf{e}_\alpha$ along β and γ directions, respectively. Note that the above equations are retrieved with the BGK second-order model with errors of $O(\mathrm{Ma}^3)$, which is a desirable simplification to reduce computational efforts in simulating incompressible flows. It can be shown that Eqs. (4) and (5) are identical to the Navier-Stokes equations with the following relationship between the single relaxation parameter τ and the kinematic viscosity υ:

$$\upsilon = c_s^2\left(\tau - \frac{1}{2}\right)\delta_t. \tag{6}$$

C-E analysis also gives the following relationship between the non-equilibrium distribution function ${f_\alpha}^{neq}$ and the equilibrium distribution function:

$$f_\alpha^{neq} = f_\alpha - f_\alpha^{eq} = -\tau\delta_t\left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha\cdot\nabla\right)f_\alpha^{eq} = -\tau\delta_t \mathrm{D} f_\alpha^{eq}. \tag{7}$$

B. Simplified and highly stable lattice Boltzmann method (SHSLBM)

Apart from its popularity in the Computational Fluid Dynamics (CFD) community, several drawbacks of the lattice Boltzmann method (LBM) are known. These drawbacks include high cost in virtual memory, inconvenient implementation of physical boundary conditions, tie-up of the streaming distance and time step, and poor numerical stability at high Reynolds number. Efforts have been continuously made to overcome these defects. One interesting solution is to evolve the macroscopic variables rather than the distribution functions. Representative models in this direction of exploration include the lattice kinetic model,[37] the link-wise artificial compressibility method (ACM),[38] the cascade lattice Boltzmann method,[39] etc.

A recent attempt in the similar direction was made by Chen *et al.*[27–29] who developed a simplified and highly stable lattice Boltzmann method (SHSLBM), which is also free of evolution of distribution functions. The basic idea of SHSLBM is to reconstruct solutions to the macroscopic governing equations [Eqs. (4) and (5)] recovered from the lattice Boltzmann equation (LBE) and resolved in a predictor-corrector step under the framework of LBM. The reconstruction process adopts the lattice properties, conservation laws, and the relationships given by C-E analysis. The resultant formulations of SHSLBM read as per Eqs. (8)–(12) below:

Predictor step:

$$\rho^* = \sum_\alpha f_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, t - \delta_t), \tag{8}$$

$$\rho^* \mathbf{u}^* = \sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, t - \delta_t). \tag{9}$$

Corrector step:

$$\rho(\mathbf{r}, t) = \rho^*, \tag{10}$$

$$\begin{aligned}\rho(\mathbf{r}, t)\mathbf{u}(\mathbf{r}, t) = \rho^* \mathbf{u}^* - \left(1 - \frac{1}{\tau}\right) \sum_\alpha \mathbf{e}_\alpha \\ \times \left[f_\alpha^{neq}(\mathbf{r} + 0.5\mathbf{e}_\alpha \delta_t, t - 0.5\delta_t)\right. \\ \left. - f_\alpha^{neq}(\mathbf{r} - 0.5\mathbf{e}_\alpha \delta_t, t - 0.5\delta_t)\right],\end{aligned} \tag{11}$$

where the non-equilibrium distribution functions are evaluated through

$$\begin{aligned}f_\alpha^{neq}(\mathbf{r} - 0.5\mathbf{e}_\alpha \delta_t, t - 0.5\delta_t) \\ = -\tau\left[f_\alpha^{eq}(\boldsymbol{r}, t) - f_\alpha^{eq}(\mathbf{r} - \mathbf{e}_\alpha \delta_t, t - \delta_t)\right].\end{aligned} \tag{12}$$

SHSLBM inherits the simplicity and explicitness of conventional LBM, while also having various advantages over it. The macroscopic variables are directly evolved in SHSLBM. Therefore, it saves a lot of virtual memory and the physical boundary conditions can be directly implemented without tedious transformation into conditions for the distribution functions as was required in conventional LBM. Moreover, through von Neumann stability analysis, SHSLBM can be theoretically proven to be more stable than conventional LBM,[29] which makes it more competitive than conventional LBM in real engineering problems with high Reynolds numbers.

The above formulations of SHSLBM can be flexibly extended to non-uniform meshes or body-fitted meshes. To realize that, an interpolation algorithm is necessary to evaluate physical properties at virtual streaming nodes around the mesh point. Numerical solutions on the virtual streaming nodes do not need to be stored. And the locations of the virtual streaming nodes are determined by the streaming distance which is usually chosen as the minimal mesh spacing.

III. HIGHLY ACCURATE SIMPLIFIED LATTICE BOLTZMANN METHOD (HSLBM)

Although SHSLBM performs very well in terms of numerical stability, memory cost, and implementation of physical boundary conditions, one main constraint is its spatial accuracy. SHSLBM can only ensure the second-order of accuracy in space, due to which fine meshes need to be adopted when dealing with problems at high Reynolds numbers. To a certain extent, such constraint counteracts the advantage brought by its stability performance. Although SHSLBM can give converged and smooth results at coarse meshes, these results are not accurate enough in high Reynolds number scenarios. To increase the accuracy on coarse meshes, a numerical scheme in a higher order of accuracy is the only option in numerical simulations. The motivation of our work is to develop a highly accurate scheme that is based on the SHSLBM algorithm. The new method should have a higher order of accuracy while maintaining the positive features of SHSLBM.

Constructing a highly accurate scheme within the conventional lattice Boltzmann framework is mathematically challenging. Starting from the continuous Boltzmann equation, improving accuracy in the discretization of the velocity space can sacrifice the explicitness of the method, while using higher-order polynomials in the equilibrium distribution function complicates the formulation.[32,33] The present highly accurate simplified lattice Boltzmann method (HSLBM) explores an alternative way to increase the overall accuracy by locally applying the formulations of SHSLBM. The basic idea of HSLBM is to distribute virtual streaming nodes to decouple the streaming distance δ_x from the mesh spacing h. The overall order of accuracy of a method is evaluated by the power of mesh spacing that its error can reach. If the streaming distance is much smaller than the mesh spacing, even though the scheme is in the second order in terms of δ_x, higher overall accuracy can be expected because it is assessed in terms of h.

The key issue in HSLBM is the correlation between the streaming distance and the mesh spacing. Since they are not equal, the streaming nodes are no longer located on the mesh points. Therefore, physical properties on the streaming nodes need to be interpolated from known values on mesh points. In order not to sacrifice the overall accuracy, a high-order interpolation algorithm should be adopted here. For regular interpolation methods, this means that more interpolation points should be used.

A. High-order interpolation algorithm

In a two-dimensional scenario with structured Cartesian mesh (see Fig. 1), the Lagrange interpolation algorithm can be adopted to interpolate the functional value at an arbitrary location (x, y). In Lagrange interpolation, n interpolation points can theoretically ensure the $(n-1)$th order of accuracy.

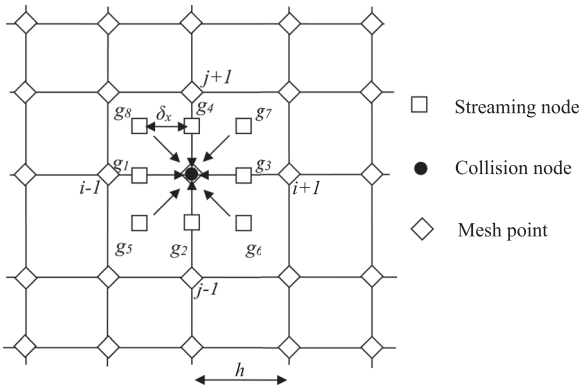


FIG. 1. Illustration of HSLBM with the D2Q9 model.

Such characteristic endows it with greater freedom to control the order of accuracy. Two-dimensional Lagrange interpolating formulation reads

$$g(x,y)=\sum_{i=1}^{M}\sum_{j=1}^{N}g\left(x_i,y_j\right)a_i(x)b_j(y), \tag{13}$$

where the Lagrange basis polynomials a_i and b_j are given by

$$a_i=\prod_{l=1,\,l\neq i}^{M}\frac{x-x_l}{x_i-x_l}, \tag{14}$$

$$b_j=\prod_{l=1,\,l\neq j}^{N}\frac{y-y_l}{y_j-y_l}. \tag{15}$$

The Lagrange basis polynomials are all functions of the relative location between the interpolation point and the mesh points, which do not vary once the streaming distance is determined. Therefore, to lower computational costs, they can be precomputed and stored before the iteration starts.

B. Formulations of HSLBM

The basic formulations of the highly accurate simplified lattice Boltzmann method (HSLBM) are similar to those of the SHSLBM. The macroscopic variables at a mesh point are updated through a predictor-corrector scheme

Predictor step:

$$\rho^{*}=\sum_{\alpha}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t), \tag{16}$$

$$\rho^{*}\mathbf{u}^{*}=\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t). \tag{17}$$

Corrector step:

$$\rho(\mathbf{r},t)=\rho^{*}, \tag{18}$$

$$\begin{aligned}\rho(\mathbf{r},t)\mathbf{u}(\mathbf{r},t)=\rho^{*}\mathbf{u}^{*}-\left(1-\frac{1}{\tau}\right)\sum_{\alpha}\mathbf{e}_{\alpha}\\ \times\left[f_{\alpha}^{neq}(\mathbf{r}+0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)\right.\\ \left.-f_{\alpha}^{neq}(\mathbf{r}-0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)\right].\end{aligned} \tag{19}$$

Note that the difference here is that, although the streaming distance δ_x is still tied up with the time step δ_t, it is no longer equal to the mesh spacing h (see Fig. 1). Such treatment is equal to introducing a set of virtual streaming nodes around the mesh point. Physical properties on the virtual streaming nodes are obtained through the high-order interpolation algorithm, while no numerical solution needs to be stored on the virtual streaming nodes.

The equilibrium distribution functions at streaming nodes $(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)$ in the predictor step are interpolated from the functional values on the surrounding mesh points (see Fig. 1), which reads

$$\begin{aligned}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)=\sum_{i=1}^{n}\sum_{j=1}^{n}f_{\alpha}^{eq}\left(\mathbf{r}_{ij},t-\delta_t\right)a_i(x-e_{\alpha x}\delta_t)\\ \times b_j\left(y-e_{\alpha y}\delta_t\right),\end{aligned} \tag{20}$$

where $\mathbf{r}=(x,y)$, $\mathbf{e}_{\alpha}=(e_{\alpha x},e_{\alpha y})$; and i and j are the index of neighboring mesh points along x- and y-directions, respectively. The non-equilibrium distribution functions are then evaluated through

$$\begin{aligned}&f_{\alpha}^{neq}(\mathbf{r}-0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)\\ &\quad=-\tau\delta_t Df_{\alpha}^{eq}(\mathbf{r}-0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)\\ &\quad=-\tau\left[f_{\alpha}^{eq}(\boldsymbol{r},t)-f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)\right]+O\left(\delta_t^3\right).\end{aligned} \tag{21}$$

Note that the above formulation approximates the non-equilibrium distribution function with the third-order of accuracy in space. As a predictor-corrector scheme, intermediate macroscopic variables obtained in the predictor step can be used to evaluate the equilibrium distribution function at the time level of t in the corrector step. Together with the equilibrium distribution function $f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)$ obtained in the predictor step, non-equilibrium distribution functions at $(\mathbf{r}-0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)$ can be directly evaluated through Eq. (21). The non-equilibrium distribution functions at $(\mathbf{r}+0.5\mathbf{e}_{\alpha}\delta_t,t-0.5\delta_t)$ are interpolated from the surrounding interpolation points shifted from the mesh points by $-0.5\mathbf{e}_{\alpha}\delta_t$. Therefore, the relative locations between the interpolation points and the interpolated point are still $\mathbf{e}_{\alpha}\delta_t$, due to which the Lagrange basis polynomials are the same as those adopted in the predictor step.

As mentioned above, the basic idea of HSLBM is to locally reconstruct the lattice model with small streaming distance on a relatively coarse mesh. The correlation between the mesh spacing and the streaming distance is accomplished by the high-order Lagrange interpolation algorithm. The numerical error of evolution can be either locally expressed as $O({\delta_x}^2)$ or globally written as $O(h^k)$, i.e., $E_{evo}=a{\delta_x}^2=bh^k$. Since a and b are two parameters in the same order of $O(1)$, adjusting the ratio between the streaming distance and the mesh spacing can alter the overall accuracy of HSLBM. In the extreme case that the streaming distance is set equal to the mesh spacing, HSLBM degrades to SHSLBM that has the second-order of accuracy. The other contributor to the overall accuracy is the interpolation accuracy, which constrains the upper limit of the accuracy of HSLBM. No matter how small the streaming distance is adopted, the overall accuracy of HSLBM cannot exceed the accuracy of the interpolation algorithm. Such characteristics will be validated in Sec. IV.

Various merits can be expected in HSLBM. On one hand, it inherits advantages of SHSLBM. The macroscopic variables, rather than the distribution functions, are directly updated during the computational process, which lowers the cost in virtual

memory and facilitates the implementation of physical boundary conditions. Since SHSLBM itself is a highly stable scheme, HSLBM is likely to maintain good numerical stability if the interpolation points and the streaming distance are properly chosen. Such behavior will be reflected in our numerical examples in high Reynolds number scenarios on very coarse meshes (see examples in Sec. V B). On the other hand, HSLBM can increase the overall order of accuracy of SHSLBM by introducing simple interpolation algorithms. This characteristic makes it possible to obtain appealing results on coarser meshes, and at the same time, maintains the simplicity of the formulations. In addition, the decoupling of the streaming distance from the mesh spacing allows more flexible selection of the time interval, which facilitates capturing high-order statistics in unsteady flows.

C. Implementation of boundary conditions

In HSLBM, the macroscopic variables are directly updated in the computational process. Therefore, the physical boundary conditions can be directly implemented without tedious transformation into conditions for the distribution functions as was required in conventional LBM. The remaining issue is how to evaluate boundary values of the non-equilibrium distribution functions that are used in the corrector step.

One popular treatment in conventional LBM is the non-equilibrium extrapolation scheme.[40] In this treatment, the non-equilibrium distribution functions on boundary mesh points are set equal to the values in the inner layer of grid points. Given the fact that the non-equilibrium distribution function is in the order of δ_x, the non-equilibrium extrapolation scheme ensures the second-order of accuracy.[40] However, such an accuracy order in boundary does not match the overall accuracy of HSLBM in which a higher order of accuracy is expected. To alleviate such constrain, the linear extrapolation of the non-equilibrium term, which was proposed by Chen *et al.*[41] and was claimed to have the third-order of accuracy, is adopted here. Note that this boundary treatment is inspired by the linear extrapolation scheme proposed by Chen *et al.*[42] but does not require the outer layer of virtual mesh points. Moreover, it will be shown in Sec. IV that the expected spatial accuracy of HSLBM is around the third order. Therefore, the linear extrapolation scheme fits well with the flow solver in terms of numerical accuracy. Specifically, the formulation of this boundary treatment on uniform mesh can be written as

$$f_\alpha^{neq}(x_b) = 2f_\alpha^{neq}\left(x_{f1}\right) - f_\alpha^{neq}\left(x_{f2}\right), \tag{22}$$

where the subscripts b, f_1, and f_2 denote the boundary node, the first, and the second inner layers of mesh points, respectively.

D. Computational sequence

The computational sequence of HSLBM can be summarized as follows:

(1) Specify the streaming distance δ_x ($\delta_x = \delta_t$). Determine the single relaxation parameter τ.
(2) Predictor step: Use the 5-point Lagrange interpolation algorithm [Eq. (13)] to determine the equilibrium distribution functions at the streaming nodes. Then employ Eqs. (16) and (17) to calculate the intermediate physical properties ρ^* and $\mathbf{u}^*$.
(3) Compute the non-equilibrium distribution function by applying Eq. (21).
(4) Implement the linear extrapolation algorithm to determine the boundary values of non-equilibrium distribution functions.
(5) Corrector step: Use the 5-point Lagrange interpolation algorithm to interpolate non-equilibrium distribution functions at $(\mathbf{r} + 0.5\mathbf{e}_\alpha\delta_t,\ t - 0.5\delta_t)$. Obtain the density and the velocity at the mesh point and new time step from Eqs. (18) and (19).
(6) Implement appropriate boundary conditions for the macroscopic physical properties.
(7) Repeat steps (2)–(6) until the computation converges or the prescribed maximum computational time is reached.

IV. VALIDATION AND OPTIMIZATION

A. Validation of spatial accuracy

Here we use the example of Taylor-Green vortex flow[43] to validate the spatial accuracy of the highly accurate simplified lattice Boltzmann method (HSLBM). Theoretical solutions to this problem are

$$u(x,\ y,\ t) = -U_0\cos(\pi x/L)\sin(\pi y/L)\exp\left(-\frac{2\pi^2 U_0 t}{\mathrm{Re}\ L}\right), \tag{23}$$

$$v(x,\ y,\ t) = U_0\sin(\pi x/L)\cos(\pi y/L)\exp\left(-\frac{2\pi^2 U_0 t}{\mathrm{Re}\ L}\right), \tag{24}$$

$$p(x,\ y,\ t) = p_0 - \frac{\rho_0 U_0^2}{4}\left[\cos(2\pi x/L) + \sin(2\pi y/L)\right] \times \exp\left(-\frac{4\pi^2 U_0 t}{\mathrm{Re}\ L}\right), \tag{25}$$

where U_0 and ρ_0 are the initial velocity amplitude and the reference density and p_0 is the background pressure. Reynolds number, which is defined as $\mathrm{Re} = U_0 L/\upsilon$, is set to 10. Periodic boundary condition is implemented on all boundaries. Solutions of the velocity at the non-dimensional time $t^* = t(U_0/L) = 2.0$ are extracted to compute the L_2 norm of its relative error following the formulation:

$$L_2 = \sqrt{\frac{\sum_{k=1}^{N\times N}\left(\frac{u_{\mathrm{numerical}} - u_{\mathrm{exact}}}{U_0}\right)^2}{N\times N}}, \tag{26}$$

where $u_{\mathrm{numerical}}$ and u_{exact} refer to the numerical result and the analytical solution, respectively.

For illustrative purposes, here we set the number of interpolation points (n) equal to 5. Theoretically, 5 point-Lagrangian interpolation has the 4th order of accuracy in space. Therefore, by adjusting the streaming distance δ_x, the overall accuracy of HSLBM is between the 2nd order and the 4th order.

Figure 2 presents the results of relative error at different mesh spacing and streaming distances. The slope k of the linearly fitted line in log scale represents the order of spatial accuracy under a given condition. It can be seen in this

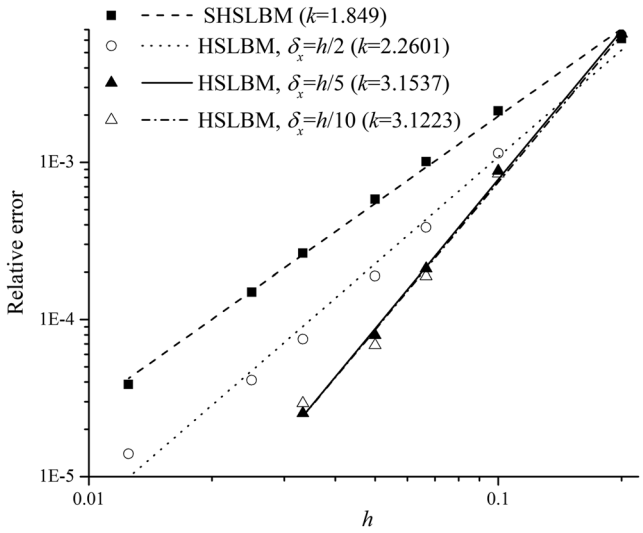


FIG. 2. L2 norms of relative error of u versus h in Taylor-Green vortex flow (n = 5).

figure that the order of accuracy increases by decreasing the ratio between streaming distance and mesh spacing. When the streaming distance is smaller than 1/5 of the mesh spacing, the order of accuracy generally converges to the 3rd order. So in practice, HSLBM with 5 interpolation points can manage to achieve the 3rd order of accuracy in space.

B. The optimal values of interpolation points and streaming distance δ_x

Apparently, the number of interpolation points and the choice of streaming distance are two major factors that affect the overall accuracy of HSLBM. To determine the optimal values of the number of interpolation points n and the streaming distance δ_x, here we still use the example of Taylor-Green vortex flow to perform comparative studies.

Two notable issues should be considered when deciding the number of interpolation points n. First, whether n is even or odd can affect the distribution of the interpolation points. The distribution of the interpolation points in turn influences the numerical stability. If n is an odd number, the interpolation points are symmetrically distributed; if n is even, then unsymmetrical interpolation is encountered, which may lead to numerical instability. The other issue is that the number of interpolation points constrains the highest order of accuracy that HSLBM can achieve in theory. Small n may yield an insufficient order of accuracy in interpolation, while large n favors the accuracy but could induce numerical stability issues.

For comprehensive investigation, five different values of n correlated with three different streaming distances are tested on the Taylor-Green vortex flow. The slope of the linearly fitted line that represents the order of accuracy is tabulated in Table I. It reveals that, in cases with even n, the computation can easily diverge when the streaming distance is smaller than 1/2 of the mesh spacing. Such instability, as mentioned above, is triggered by the unsymmetrical distribution of the interpolation points. Even in cases with $\delta_x = h/2$ that the interpolation can work, the resultant order of accuracy in cases with even n is still lower than the results obtained by $n = 5$ in the same condition (see Fig. 3). Therefore, it is suggested to use the odd number of interpolation points in computations.

TABLE I. Slopes of the linearly fitted line at different cases ("...” means the computation diverges in that case).

	$\delta_x = h/2$	$\delta_x = h/5$	$\delta_x = h/10$
$n = 3$	1.8012	1.7093	1.6390
$n = 4$	2.1576	...	...
$n = 5$	2.2601	3.1537	3.1223
$n = 6$	1.8622	...	...
$n = 7$	2.0112	3.1399	4.4585

Among all the odd numbers of n in Table I, the practical order of accuracy in cases with $n = 3$ is always lower than 2nd order. This is because the theoretical interpolation accuracy in this condition is 2nd order. Thus, no matter how the streaming distance is adjusted, the overall accuracy is constrained by the interpolation accuracy and cannot reach the 2nd order. Cases with $n = 5$ and $n = 7$ reveal a similar order of accuracy when streaming distances are set to be $\delta_x = h/2$ or $h/5$. It is quite interesting that if the streaming distance is shortened to $\delta_x = h/10$, using 7 interpolation points can achieve more than the 4th order of accuracy ($k = 4.4585$). However, one should be reminded that adopting so many interpolation points (equivalent to adopting 6th order polynomials) can trigger numerical instability, especially in cases where no-slip boundary is encountered and the interpolation points are unsymmetrically distributed near the solid boundary. Indeed, in the test of plane Poiseuille flow (see Sec. V A for problem setups) with the same settings ($n = 7$, $\delta_x = h/10$) on 11 mesh grids, the computation diverges. On the contrary, with $n = 5$ and $\delta_x = h/5$, very good agreement with the analytical solution can be achieved (see Fig. 10 in Sec. V A).

To sum up, the evaluation of the optimal values of the number of interpolation points and the streaming distance is based on the balance and the compromise among numerical accuracy, stability, and efficiency. Increasing the number of interpolation points while decreasing the streaming distance contributes to the numerical accuracy in theory. However, it is

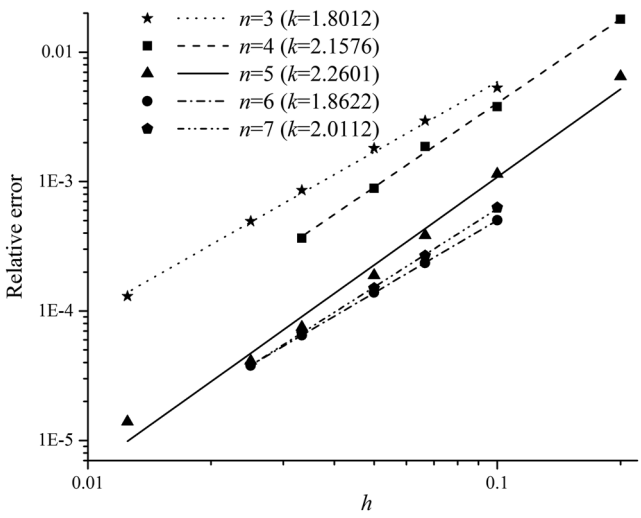


FIG. 3. Relative error vs. mesh spacing: $\delta_x = h/2$.

found that $n = 7$ can lead to stability issues in practical implementations, while $n = 3$ cannot achieve the desired order of accuracy. Therefore, the optimal value of the number of interpolation points is $n = 5$. In such a condition, the comparative study indicates that the order of accuracy does not vary much when the streaming distance is less than $\delta_x = h/5$. Since the streaming distance and the time step are tied up in the LBM framework, small streaming distance yields small time step and in turn increases computational cost. Thus, 5 interpolation points and the streaming distance of $\delta_x = h/5$ will be adopted, and overall the third order of accuracy can be expected.

C. Validation of temporal accuracy

The simplified and highly stable lattice Boltzmann method (SHSLBM) can ensure the second-order of accuracy both in space and in time when reconstructing macroscopic variables.[29,41] Since HSLBM uses the same evolution formulation as SHSLBM with an extra interpolation algorithm that can only affect the spatial accuracy, theoretically HSLBM should also preserve the second-order of accuracy in time. In this part, the numerical test is carried out in the example of Taylor-Green vortex flow to investigate the practical order of temporal accuracy of HSLBM. A uniform mesh size of 41×41 is employed, and the 5-point Lagrange interpolation algorithm is adopted. Note that in the present LB model, the time step is tied-up with the streaming distance. Therefore, adjusting the streaming distance δ_x is equivalent to altering the time interval δ_t.

Relative errors at different time intervals are shown in Fig. 4. Also included in this figure is the linearly fitted line in log scale. The slope of the fitted line reads $k = 1.811$, which indicates that approximately the second-order of temporal accuracy can be achieved by HSLBM.

D. Stability and mass/energy conservation

To further validate the proposed HSLBM, we present numerical tests on the stability and mass/energy conservation in this subsection. The above Taylor-Green vortex flow is still utilized as an example. A uniform mesh size of 81×81 is adopted. The relaxation parameter is set to be $\tau = 0.5$ as computation commences, which corresponds to the inviscid condition.[44–46] A long-term numerical test under the above setups allows a glance at the performance of HSLBM on numerical stability and mass/energy conservation. On one hand, it is generally believed that the conventional LB model is highly unstable as the relaxation parameter approaching 0.5. Therefore, the above settings are extremely likely to induce numerical instability for LB models. On the other hand, given the inviscid characteristic of the flow, the velocity should not decay in physics. Thus, the numerical test should fulfill both the mass conservation and the conservation of kinetic energy to maintain consistency.

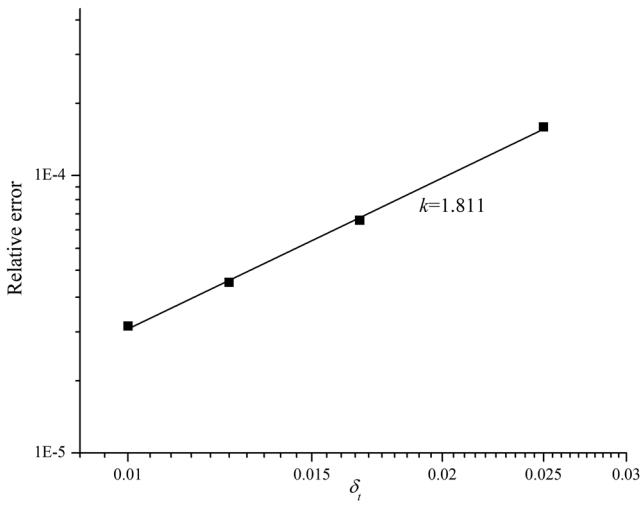


FIG. 4. Relative error vs. time interval. Mesh size: 41×41; n = 5.

Figure 5 presents the velocity contours obtained by HSLBM at different times. Two important notions can be inferred from these contours. First, stable and smooth computational results are obtained in the long-term simulation, which indicates the good numerical stability of HSLBM. Second, the magnitude of velocity maintains constant as time marches. Such typical inviscid characteristic is successfully captured by HSLBM, which validates its low numerical dissipation and demonstrates its consistency with the physical problem.

Time histories of mass and kinetic energy are depicted in Fig. 6 to shed light on the conservation performance. More details are presented in Fig. 7 which illustrates the time histories of relative errors of the velocity, total mass, and kinetic energy. As can be seen in these figures, the variation of mass and kinetic energy is maintained within 0.2% in the long-term simulation. Given their small amplitudes, these outcomes prove that HSLBM effectively controls the numerical diffusion and preserves the nice features of mass and energy conservations.

E. Validation of spatial accuracy on body-fitted mesh

In practice, curved boundaries are often encountered, for which the body-fitted mesh is preferred to be used. To reveal the spatial accuracy of the proposed method on body-fitted meshes, we resolve the same problem of Taylor-Green vortex flow on the computational domain of concentric annulus (see Fig. 8). The annulus is centered at the origin of the coordinate and its inner and outer radii are 1 and 2, respectively. Analytical solutions to the flow field are directly assigned on boundary mesh points, while the governing equations are resolved on inner mesh points to update physical quantities defined on them. The mesh points are uniformly distributed along the radial and the azimuthal directions. The radial spacing is selected as the characteristic mesh spacing h, while the maximum azimuthal mesh spacing is ensured to be smaller than the radial spacing. Computational results of the test are presented in Fig. 9 in log scale. The linearly fitted line reveals the slope of 3.012, which is very close to the previous results obtained on uniform meshes in the Cartesian coordinate system. The above test indicates that the proposed HSLBM can also achieve the third-order of spatial accuracy on body-fitted meshes.

V. NUMERICAL EXAMPLES

For comprehensive evaluation of the robustness of the highly accurate simplified lattice Boltzmann method

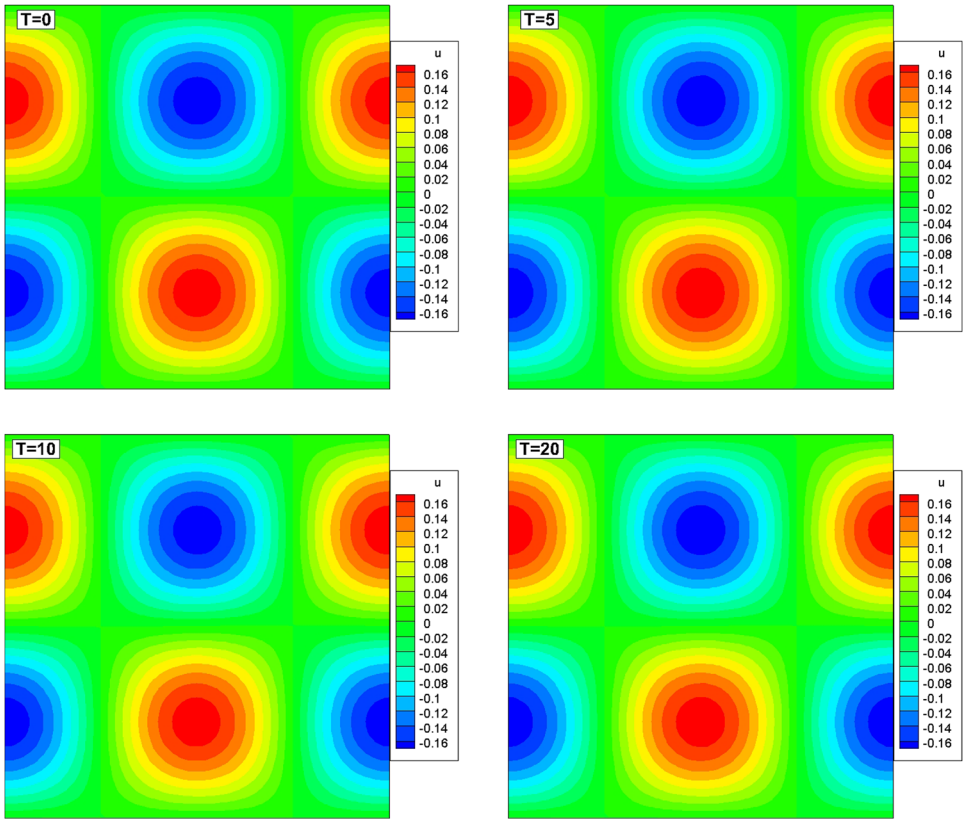


FIG. 5. Velocity contours of inviscid simulation of the Taylor-Green vortex at T = 0, 5, 10, and 20.

(HSLBM), five representative examples are presented in this section. Comparisons are made to analytical/numerical results in literature or results given by SHSLBM with non-equilibrium extrapolation scheme as its boundary treatment.

A. Plane Poiseuille flow

In the first example, the classic plane Poiseuille flow is simulated to examine the robustness of HSLBM. The geometric configuration of this problem is a pair of parallel plates. The flow between the plates is driven by a constant pressure gradient until reaching a steady state. Analytical solution to this problem is available, in which the velocity profile is given by

$$u(y) = \frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left(y^2 - hy\right), \tag{27}$$

where the distance between two plates h is 1 and the pressure gradient is $\partial p/\partial x = -0.0001$ in our computations. The vertical mesh size is $N_y = 5$. Dynamic viscosity is set to be $\mu = 0.001$. The periodic boundary condition is implemented on the left and the right boundaries, while no-slip boundary condition is used on the upper and the bottom walls.

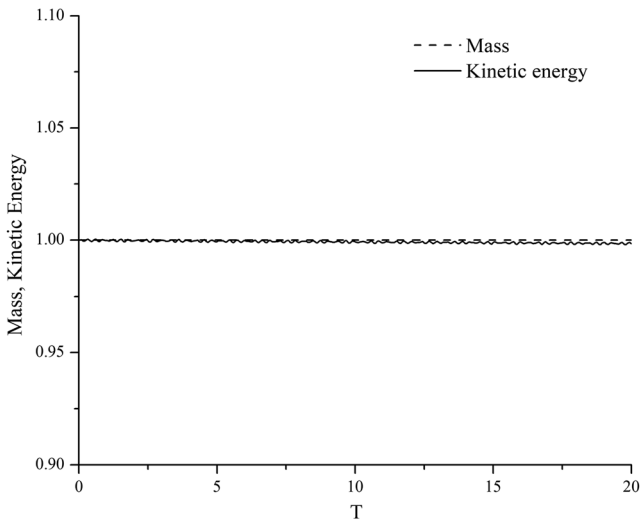


FIG. 6. Time history of the total mass and kinetic energy for the inviscid simulation of Taylor-Green vortex flow.

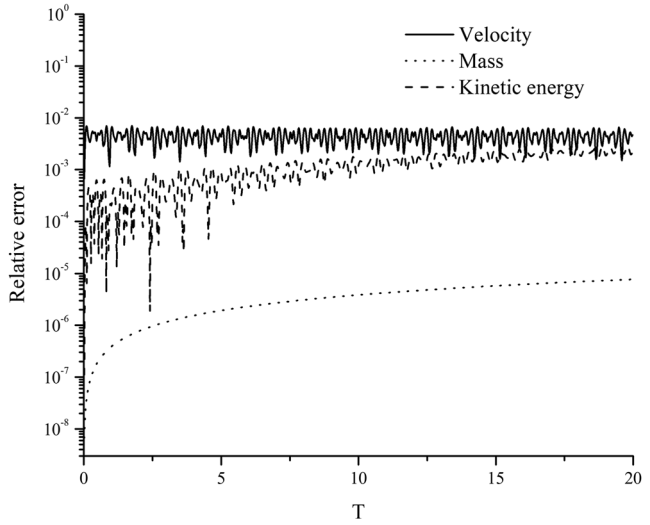


FIG. 7. Time histories of relative errors of the velocity, total mass, and kinetic energy.

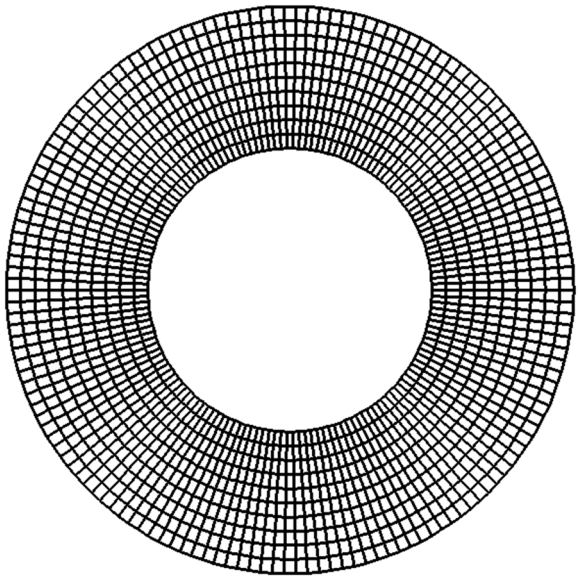


FIG. 8. Illustration of the body-fitted mesh for the Taylor-Green vortex flow.

Velocity profiles obtained by HSLBM and SHSLBM are depicted in Fig. 10. It is clearly shown in this figure that the numerical results of HSLBM perfectly match the analytical solutions, while large gaps from the analytical data are observed in the SHSLBM simulation at the same mesh resolution. Further refinement of the mesh size to $N_y = 24$ is required in SHSLBM to get comparable results. Referring to previous reports,[47,48] conventional LBM also requires more than 16 mesh points to achieve appealing agreement. Such outcomes evidently prove that HSLBM performs much better than SHSLBM in terms of numerical accuracy.

B. 2D lid-driven cavity flow

The plane Poiseuille flow is basically a simple example with one-dimensional parabolic velocity profile. To test the robustness of HSLBM in a more real problem, here we present the example of lid-driven cavity flow. The physical configuration of the problem is a square cavity with a constantly moving lid. The fluid inside the cavity is driven by

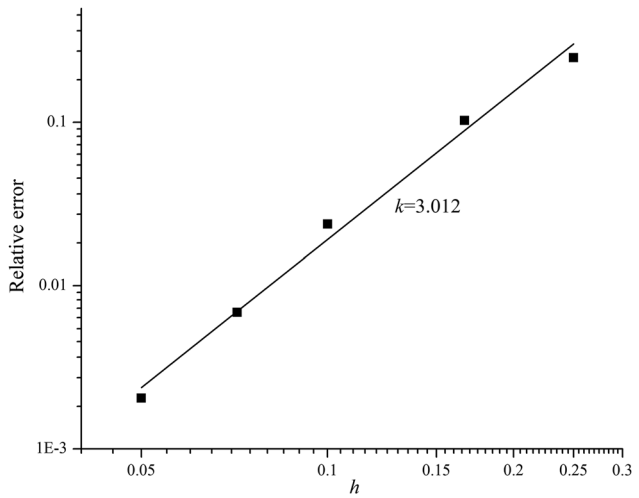


FIG. 9. Relative error vs. mesh spacing for the test on the body-fitted mesh.

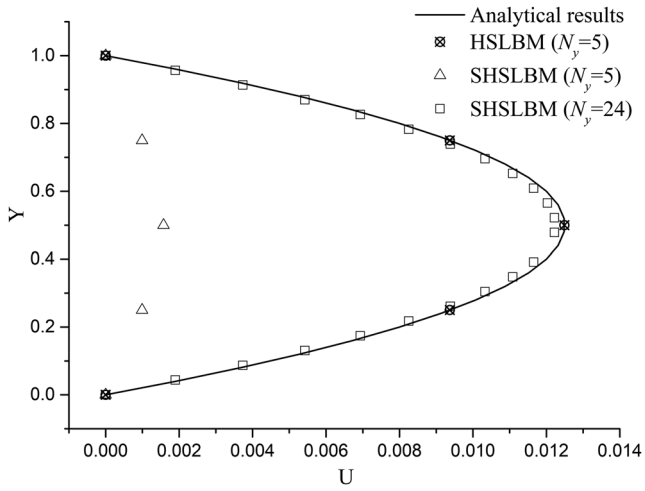


FIG. 10. Velocity profiles obtained by HSLBM and SHSLBM in plane Poiseuille flow.

the moving lid to form a circulation flow. The most important non-dimensional parameter for this problem is the Reynolds number defined as $\mathrm{Re} = U_{\mathrm{lid}}L/\upsilon$, where U_{lid}, L, and υ denote the lid velocity, the length of the cavity, and the kinematic viscosity of the fluid, respectively. When Reynolds number is lower than 10 000, it is generally believed that the viscous effect is strong enough to constrain the flow within a steady state.[49]

Our numerical tests are carried out on a uniform mesh size of 125 × 125, and five different Reynolds numbers of 400, 1000, 5000, 7500, and 10 000 are considered. Velocity profiles along the centerlines are presented in Figs. 11 and 12. Apparently, the numerical results obtained by HSLBM show better agreement than those given by SHSLBM at the same mesh size. The advantage of HSLBM is more significant in cases with higher Reynolds numbers. For example, in the case with Re = 10 000, HSLBM can achieve desirable results on a uniform mesh size of 125 × 125 for which SHSLBM needs to use a finer mesh of 301 × 301.[29] The performance matches our expectation that HSLBM has a higher order of accuracy than SHSLBM and can give more accurate results at the same mesh resolution.

The computational cost of HSLBM is tabulated in Table II. The case with Re = 10 000 is chosen for illustration here. The performance of HSLBM is compared with that of SHSLBM at the same mesh size and at the mesh size with comparable accuracy. As can be seen in the table, under the same mesh size, HSLBM consumes more virtual memory and computational time than SHSLBM. This is in accordance with our expectations because the interpolation algorithms are involved in HSLBM, and its time step is 1/5 of that in SHSLBM with the same mesh size. As mentioned above, to achieve comparable accuracy at this Reynolds number, SHSLBM requires a finer mesh size of 301 × 301. In such a condition, both the virtual memory and the computational time of SHSLBM are much larger than those of HSLBM with the mesh size of 125 × 125. Specifically, the virtual memory and the computational time of HSLBM are about 30.1% and 24.1% those of SHSLBM, respectively. The results indicate that although SHSLBM is a highly stable scheme and can get converged results on coarse

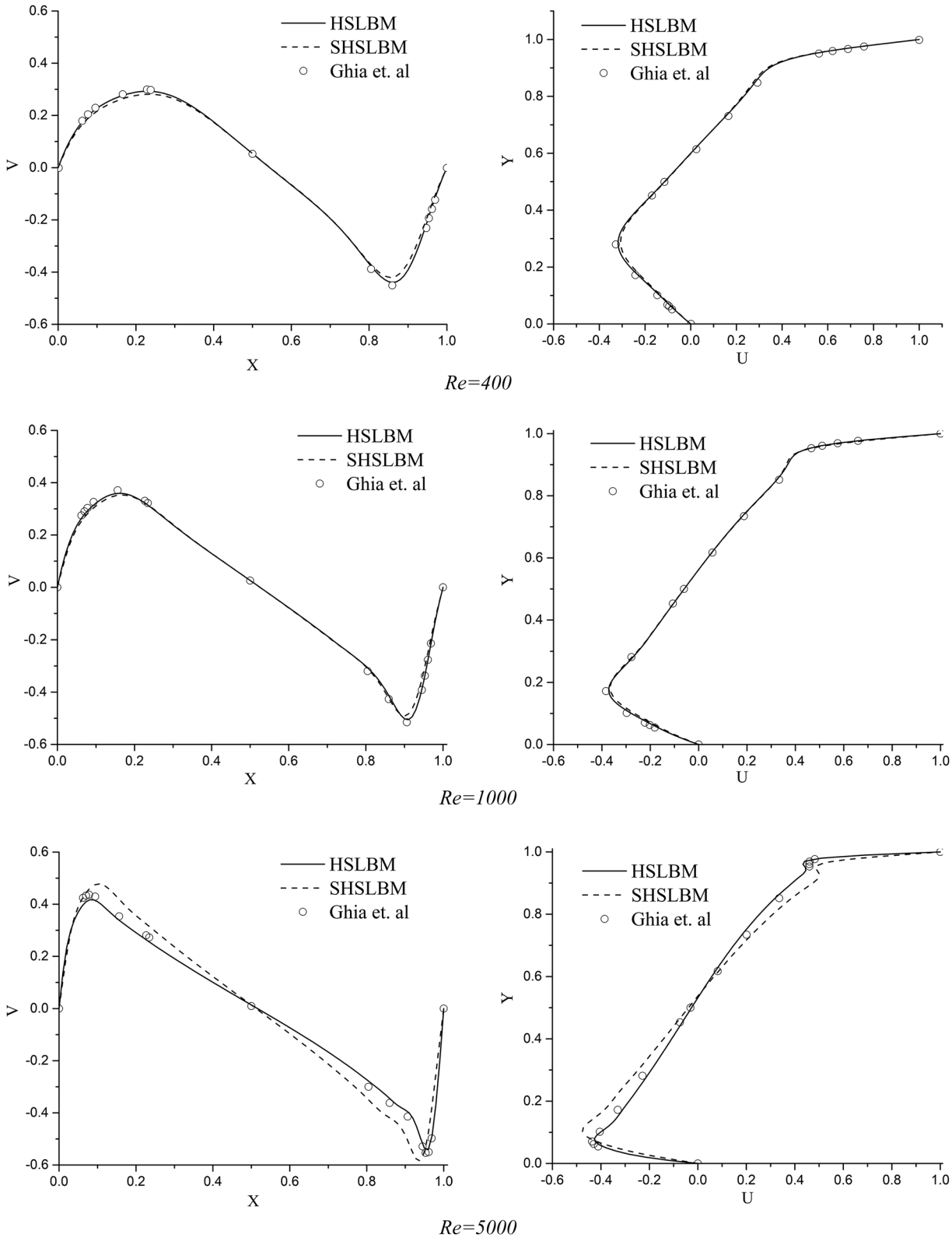


FIG. 11. Velocity profiles along the centerlines in lid-driven cavity flows at Re = 400, 1000, and 5000. Mesh size: 125 × 125.

meshes, its results are not accurate enough. To achieve comparably accurate results, HSLBM is much more efficient than SHSLBM in computational consumptions.

Another notable issue is the implementation of boundary conditions for the non-equilibrium distribution function. For problems with solid boundaries, the overall accuracy of the numerical outcome is determined by both the accuracy of the flow solver and the accuracy of the boundary condition. As discussed before, the spatial accuracy of HSLBM under recommended settings is equal to or slightly higher than the

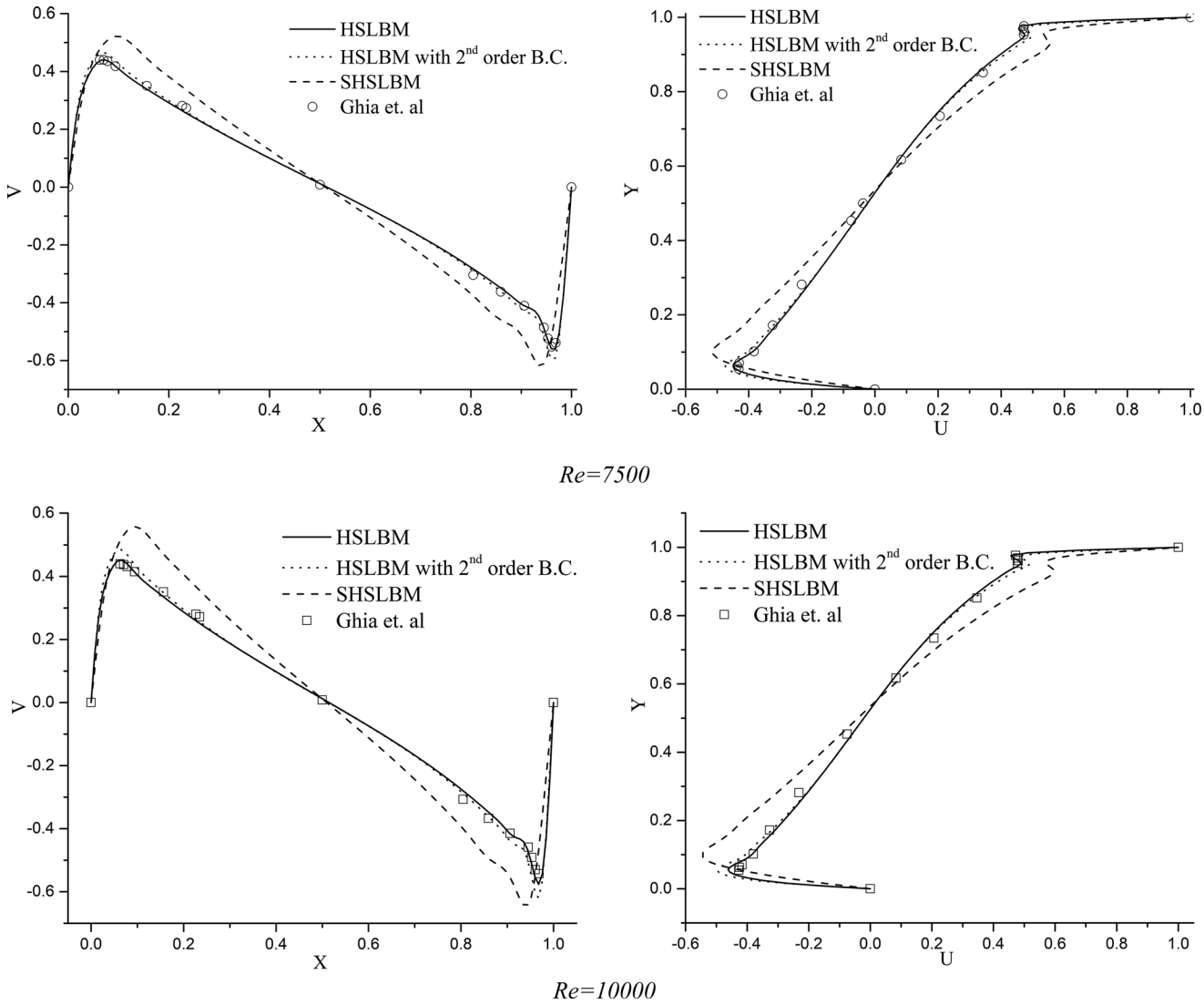


FIG. 12. Velocity profiles along the centerlines in lid-driven cavity flows at Re = 7500 and 10 000. Mesh size: 125 × 125.

third order. The normal non-equilibrium extrapolation scheme that has the second-order of accuracy (denoted as "2nd order B.C." in Fig. 12) does not fit the flow solver in terms of numerical accuracy, which may interfere the overall performance of HSLBM. Therefore, the linear extrapolation scheme with the third-order of accuracy is, in theory, a better choice when implementing boundary conditions. Such advantage is clearly reflected in Fig. 12. When adopting HSLBM as the flow solver, simply replacing the non-equilibrium extrapolation scheme with the linear extrapolation approach significantly increases the accuracy of numerical results. Another notable issue in this figure is that with the same boundary treatment, HSLBM still gives better results than SHSLBM. This phenomenon can be numerically interpreted as being favored by the higher order of accuracy in a flow solver.

TABLE II. Comparison of computational cost of HSLBM and SHSLBM in lid-driven cavity flow at Re = 10 000.

Methods	HSLBM	SHSLBM		LBM
Grid size	125 × 125	125 × 125	301 × 301	301 × 301
Virtual memory (Megabytes)	4.1	3.0	13.6	24.0
Computational time (s)	17 172	11 398	71 380	67 712

Streamlines obtained by HSLBM at Reynolds numbers of 7500 and 10 000 are presented in Fig. 13. Smoothness is well maintained in the simulation. Delicate flow structures such as the secondary vortex on the right-bottom corner are captured. Locations of the primary vortex at various Reynolds numbers of 1000, 5000, and 7500 are collected and compared with reference data[49–51] in Table III. Quantitative comparison indicates good agreement, which further validates the accuracy of HSLBM.

Finally, the numerical stability of HSLBM is tested in the case with Re = 7500 on various mesh sizes. Velocity profiles along the centerlines are depicted in Fig. 14. It turns out that, even on very coarse mesh size of 11 × 11, HSLBM can still give converged and smooth results. It is also noteworthy that the Mach number here can be lowered to less than Ma = 0.02 at the mesh size of 11 × 11 without inducing numerical

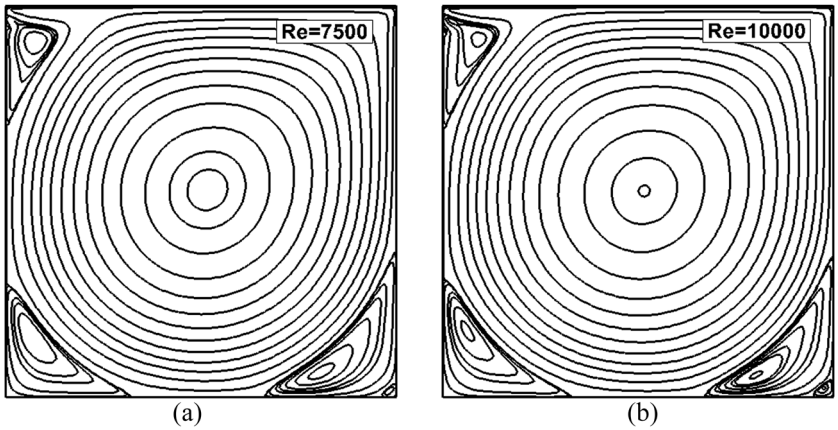


FIG. 13. Streamlines of lid-driven cavity flow at Re = 7500 and 10 000.

instability, which corresponds to a relaxation parameter smaller than $\tau = 0.5002$. In previous LB models, such small values of relaxation parameter usually lead to divergence.[50] Further test reveals that the maximum stable Reynolds number of HSLBM at the mesh size of 11×11 is higher than 10^5, which is far above previous LB models (see Fig. 5 in Ref. 52). Recalling the stability analysis performed in Ref. 29, SHSLBM is stable to all wavenumbers in high Reynolds number scenarios under the incompressible limit. The present test further validates that HSLBM inherits the nice stability features from SHSLBM. Therefore, HSLBM is suitable for real engineering problems in which a high Reynolds number is frequently encountered.

TABLE III. Comparison of the location of the primary vortex at Re = 1000, 5000, and 7500. The boldface values refer to the results obtained by the present HSLBM.

Reynolds number	References	X location	Y location
Re = 1000	Ghia *et al.*[49]	0.5313	0.5625
	Wu and Shao[51]	0.5353	0.5687
	Present	**0.5325**	**0.5657**
Re = 5000	Ghia *et al.*[49]	0.5117	0.5352
	Wu and Shao[51]	0.5164	0.5371
	Asinari and Karlin[50]	0.5221	0.5341
	Present	**0.5185**	**0.5335**
Re = 7500	Ghia *et al.*[49]	0.5117	0.5322
	Wu and Shao[51]	0.5115	0.5341
	Present	**0.5152**	**0.5308**

C. Polar cavity flow

To examine the flexibility and accuracy of HSLBM on the body-fitted mesh, the numerical test on polar cavity flow is carried out. The physical configuration is a sector which spans over the angle of $\theta = 1$ and is bounded by two curved walls at the radius of R_i and R_o (see Ref. 53 for details). The Reynolds number that governs the flow pattern is defined as $\mathrm{Re} = U_\theta(R_o - R_i)/\upsilon$, where U_θ denotes the constant azimuthal velocity in the inner boundary at radius R_i. Since the boundaries are curved in this problem, regular uniform Cartesian mesh cannot be utilized. In our simulation, the computational domain is discretized by a body-fitted mesh comprising the radial and the azimuthal directions. Interpolation algorithms are performed along the computational grid directions, i.e., via radial and azimuthal directions.

Existing experimental and numerical data are mainly in the Reynolds number of 350.[53–56] Therefore, we also perform test under this Reynolds number to validate the accuracy of HSLBM. In Fig. 16, the distributions of radial and azimuthal velocities along the centerline at $\theta = 0.5$ are presented. Specifically, the mesh size adopted in our simulation is 41×41; and the mesh points are uniformly distributed along the radial and the azimuthal directions (see Fig. 15).

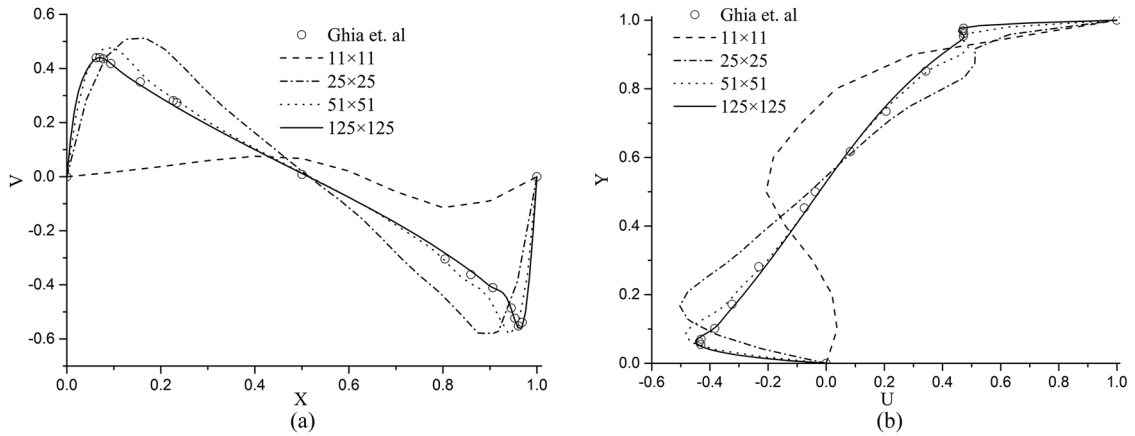


FIG. 14. Velocity profiles along the centerlines obtained by HSLBM with different mesh sizes in lid-driven cavity flow at Re = 7500.

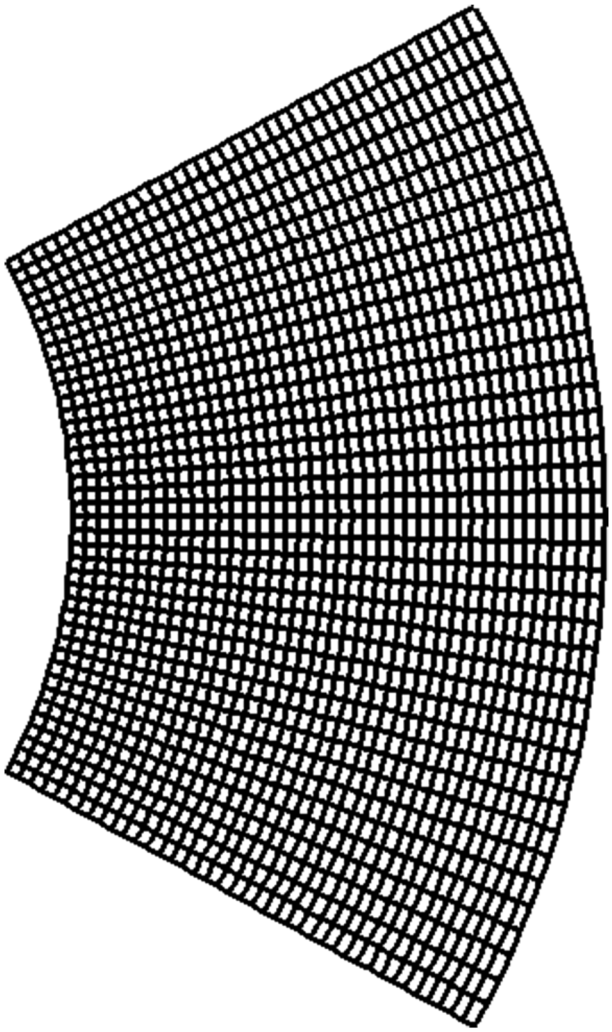


FIG. 15. Illustration of the mesh adopted in simulation of polar cavity flow (mesh size 41 × 41).

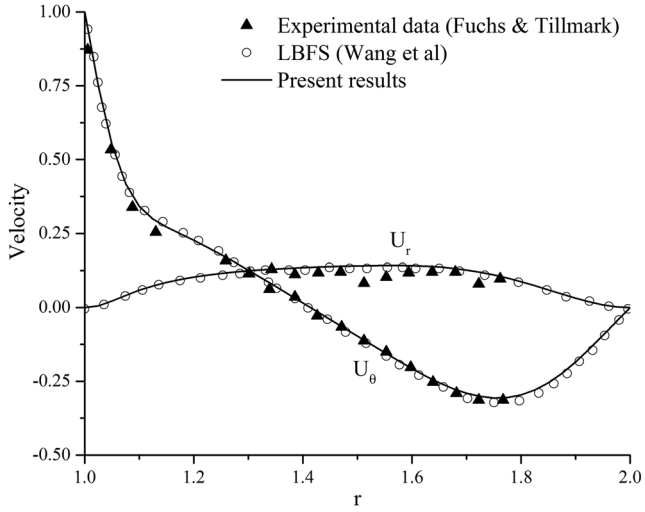


FIG. 16. Velocity distributions along radial and azimuthal directions for polar cavity flow at Re = 350.

Note that previous second-order schemes usually require 81 × 81 non-uniform mesh to achieve well converged results.[53–56] Numerical results obtained by HSLBM are compared with reference data reported in literature studies.[53,54] As can be seen in Fig. 16, good agreement is achieved in our test, which demonstrates the robustness of HSLBM in problems with curved boundaries and on body-fitted meshes.

Next, we use the validated code to perform more tests on higher Reynolds numbers. As was initially reported by Fuchs and Tillmark,[53] flow field under a higher Reynolds number becomes more complex. With the increase of Reynolds number, two minor vortices continuously grow and squeeze the primary vortex (see Figs. 9–11 in Ref. 53). Here, we want to recover such a phenomenon and give some benchmark results in numbers for future comparisons. Two Reynolds numbers of Re = 1000 and 2000 are considered here. Under these Reynolds numbers, the flow is still within the laminar region, for which steady solution exists. A series of tests on various mesh sizes are conducted; and the distributions of the radial and azimuthal velocities are depicted in Fig. 17. Under the Reynolds number of 1000, velocity distribution does not vary much when the mesh size is finer than 81 × 81. The computation generally converges at the mesh size of 101 × 101 in the case of Re = 2000. Therefore, we consider the results obtained by the mesh size of 201 × 201 as benchmark results. In Table IV, velocity distributions along the radial and azimuthal directions

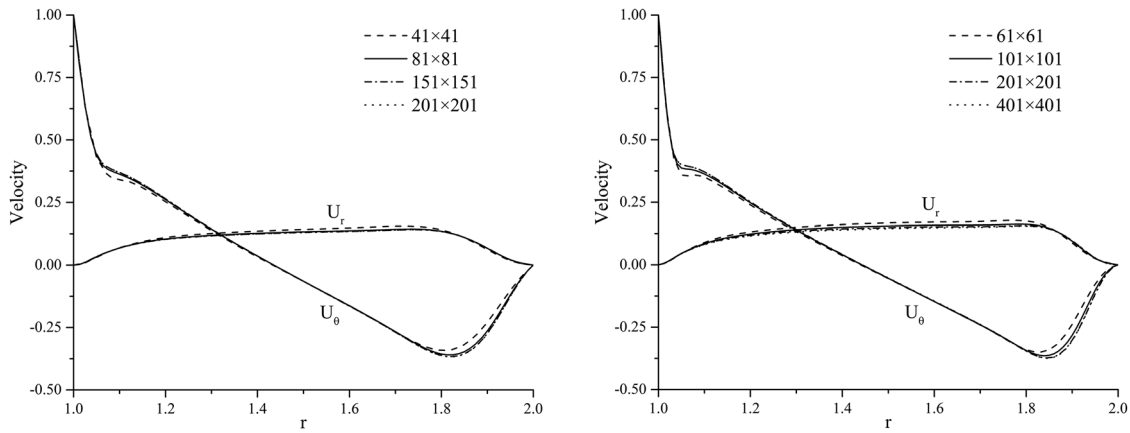


FIG. 17. Velocity distributions along radial and azimuthal directions at different mesh sizes for polar cavity flow. Left: Re = 1000; Right Re = 2000.

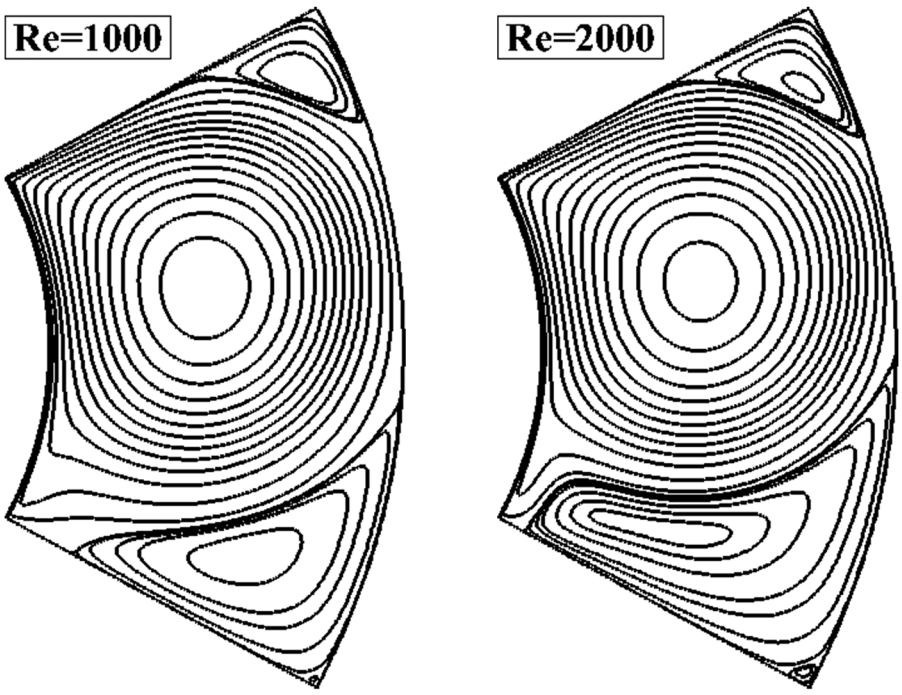
TABLE IV. Benchmark results for the velocity distributions along radial and azimuthal directions for polar cavity flow at Re = 1000 and Re = 2000.

	Re = 1000		Re = 2000	
r	U_r	U_θ	U_r	U_θ
1	0	1	0	1
1.02	0.007 44	0.704 41	0.012 47	0.584 05
1.04	0.024 76	0.498 31	0.034 81	0.417 27
1.06	0.042 1	0.413 3	0.053 81	0.395 05
1.08	0.056 34	0.385 16	0.069 46	0.386 53
1.12	0.077 76	0.353 58	0.092 79	0.350 66
1.16	0.092 63	0.312 29	0.107 9	0.301 35
1.2	0.102 64	0.264 16	0.118 23	0.251 44
1.24	0.109 46	0.214 79	0.125 97	0.205
1.28	0.114 44	0.167 04	0.132 01	0.161 66
1.32	0.118 37	0.121 61	0.136 73	0.120 37
1.36	0.121 67	0.078 17	0.140 4	0.080 45
1.4	0.124 5	0.036 18	0.143 29	0.041 49
1.44	0.126 95	−0.004 83	0.145 61	0.003 21
1.48	0.129 07	−0.045 18	0.147 5	−0.034 64
1.52	0.130 91	−0.085 07	0.149 06	−0.072 24
1.56	0.132 55	−0.124 72	0.150 33	−0.109 72
1.6	0.134 17	−0.164 44	0.151 3	−0.147 18
1.64	0.136 09	−0.204 92	0.151 93	−0.184 64
1.68	0.138 51	−0.247 13	0.152 32	−0.222 23
1.72	0.140 9	−0.291 24	0.152 99	−0.260 84
1.76	0.140 88	−0.333 41	0.154 72	−0.302 59
1.8	0.134 24	−0.362 69	0.155 88	−0.346 04
1.84	0.116 86	−0.362 65	0.148 22	−0.373 15
1.88	0.086 87	−0.317 2	0.120 23	−0.347 11
1.92	0.047 69	−0.219 82	0.068 4	−0.236 48
1.96	0.013 44	−0.095 74	0.017 4	−0.081 09
2	0	0	0	0

at $\theta = 0.5$ are tabulated for future references. The computed streamline patterns are also offered in Fig. 18. Apart from the enlarging minor vortices reported in previous studies, we find a newly emerged small vortex at the right bottom corner in cases with Re = 1000 and 2000. The size of this vortex grows with the increase of Reynolds number. This new recirculation structure is physically deduced by the smaller viscous effect at higher Reynolds number. From a numerical perspective, capturing such a delicate flow structure is not an easy task, which also indicates the accuracy of HSLBM.

D. Flow past a stationary circular cylinder

To further demonstrate the ability of the present solver in simulating unsteady flow problems with curved boundaries, validation on the example of the flow past a stationary circular cylinder is performed in this subsection. The physical configuration of this problem is a uniform flow past a stationary circular cylinder. Solutions to this problem are determined by the non-dimensional parameter of Reynolds number $\mathrm{Re} = U_{\mathrm{inf}} D/\nu$, where U_{inf}, D, and ν denote the free-stream velocity, the diameter of the cylinder, and the kinematic viscosity of the fluid, respectively. Here, special attention is paid to the unsteady flows, and two cases of Re = 100 and 200 are considered. A non-uniform mesh size of 201×301 is adopted in simulations. The mesh points are distributed uniformly along the angular direction and are stretched exponentially along the radial direction to the far field of $R_{\max} = 100D$. The free-stream velocity is set to be $U_{\mathrm{inf}} = 0.1$ to fulfill the incompressible limit. The drag coefficient C_d and the lift coefficient C_l are, respectively, defined as



(a) (b)

FIG. 18. Streamline pattern for polar cavity flow at Re = 1000 and 2000.

TABLE V. Comparison of dynamic parameters of unsteady flow past a circular cylinder at Re = 100 and 200. The boldface values refer to the results obtained by the present HSLBM.

Re	References	C_l	C_d	St
100	Braza *et al.*[60]	±0.30	1.28 ± 0.02	0.16
	Shu *et al.*[57]	±0.33	1.334 ± 0.009	0.164
	Ding *et al.*[61]	±0.28	1.325 ± 0.008	0.164
	Present	**±0.33**	**1.324 ± 0.011**	**0.163**
200	Braza *et al.*[60]	±0.78	1.38 ± 0.07	0.19
	Shu *et al.*[57]	±0.69	1.338 ± 0.045	0.197
	Ding *et al.*[61]	±0.60	1.327 ± 0.045	0.196
	Present	**±0.68**	**1.327 ± 0.048**	**0.194**

$$C_d = \frac{F_D}{\frac{1}{2}\rho U_{\text{inf}}^2}, \tag{28}$$

$$C_l = \frac{F_L}{\frac{1}{2}\rho U_{\text{inf}}^2}, \tag{29}$$

where F_D and F_L are the total drag and lift forces which are evaluated by

$$F_D = -\int_\Gamma \left[p n_x - \mu \frac{\partial u}{\partial \mathbf{n}} \right] \mathrm{d}\Gamma, \tag{30}$$

$$F_L = -\int_\Gamma \left[-p n_y + \mu \frac{\partial v}{\partial \mathbf{n}} \right] \mathrm{d}\Gamma, \tag{31}$$

where Γ represents the surface of the cylinder, n_x and n_y are the horizontal and the vertical components of the outer unit normal vector $\mathbf{n}$, and μ is the dynamic viscosity of the fluid.

Computational results of the lift coefficient, the drag coefficient, and the Strouhal number are presented in Table V. Good agreement between the present results and the reference data is observed in this table, which further validates the accuracy of the proposed method and its flexibility in resolving unsteady problems with curved boundaries. It is also noted that the mesh adopted in this test is coarser than the meshes in previous implementations with second-order solvers,[57–59] while comparable results are obtained. Such performance indicates that the present method preserves higher than the second-order of accuracy, which fulfills our expectations.

E. 3D lid-driven cavity flow

The previous examples are all two-dimensional. To further evaluate the performance of HSLBM in three-dimensional simulations, we present the example of 3D lid-driven cavity flow in this section. The physical configuration is common and can be found in many previous studies.[62–64] The key non-dimensional parameter, Reynolds number, is defined as

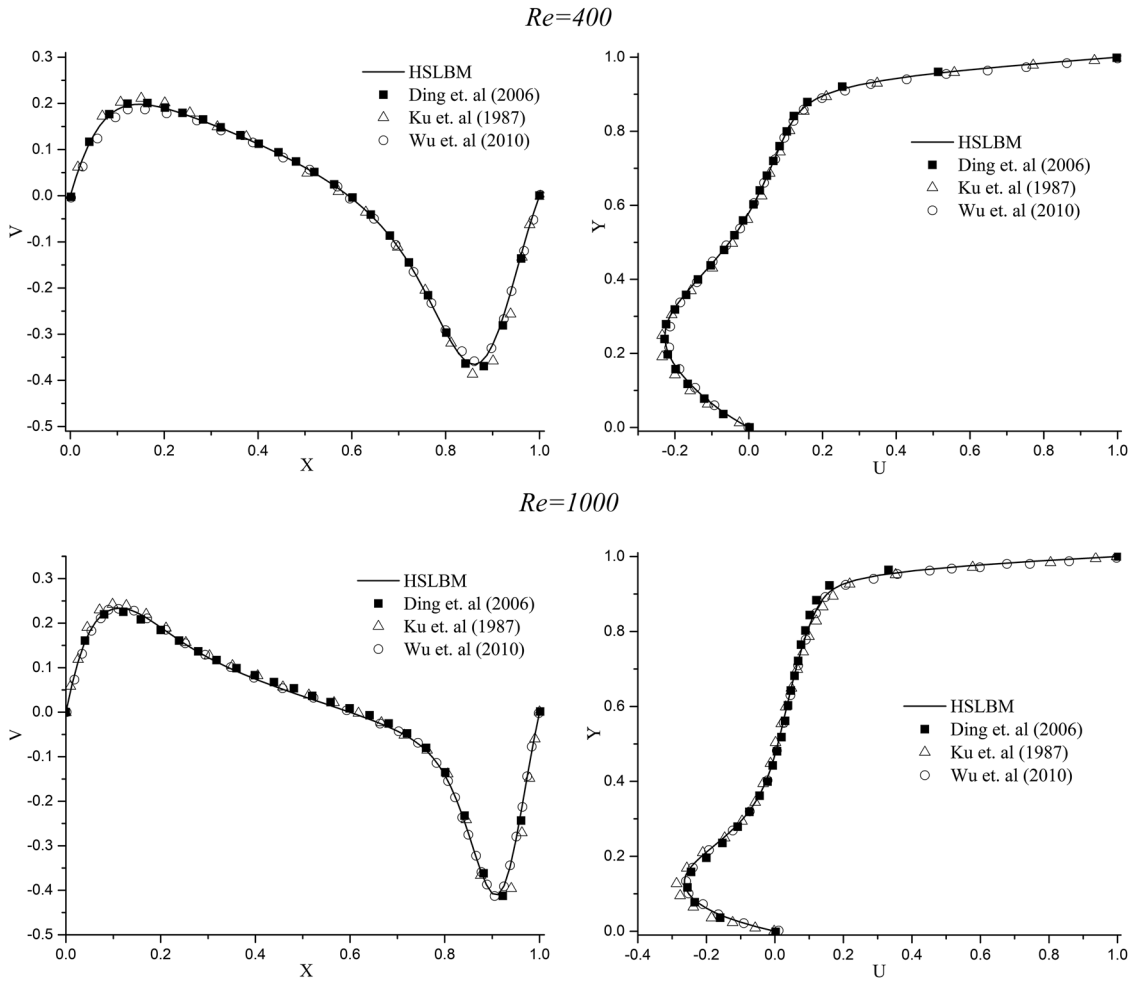


FIG. 19. Velocity distributions along centerlines in 3D lid-driven cavity flow. Left: x-V curve; Right: U-y curve.

$\mathrm{Re} = U_{\mathrm{lid}}L/\upsilon$, where U_{lid} is the constant velocity of the moving lid and L is the length of the cubic cavity. Formulations of HSLBM in three-dimensional condition are basically the same as those of two-dimensional formulations. Readers may refer to the Appendix for details.

Two cases with Reynolds numbers of 400 and 1000 are simulated by HSLBM. The uniform mesh adopted in our simulations is in the size of $81 \times 81 \times 81$. Quantitative comparisons are carried out on the velocity distribution along the centerlines. As shown in Fig. 19, the computational results obtained by HSLBM agree well with previous data in the literature.[62–64] Therefore, the robustness of HSLBM in three-dimensional problems is validated.

VI. CONCLUDING REMARKS

A highly accurate simplified lattice Boltzmann method (HSLBM) is proposed in this paper for simulations of two- and three-dimensional incompressible viscous flows. This method is developed within the framework of the simplified and highly stable lattice Boltzmann method (SHSLBM). By introducing virtual streaming nodes to decouple the streaming distance from the mesh spacing, HSLBM effectively combines the advantages of the local second-order SHSLBM scheme and the overall high order scheme. The correlation between the streaming distance and the mesh spacing is achieved by a high-order Lagrange interpolation algorithm. By adjusting various parameters in numerical experiments, it is found that using 5 interpolation points along each direction and setting streaming distance as 1/5 of the mesh spacing give optimal results. Under these settings, HSLBM can manage to achieve the third-order of spatial accuracy when solving the flow field.

To fit the third-order of accuracy of HSLBM as the flow solver, we also propose a linear extrapolation scheme for deciding boundary values of non-equilibrium distribution functions. Unlike the previous non-equilibrium extrapolation scheme, the new boundary treatment can ensure a higher order of accuracy. This expectation is also validated in the numerical test on lid-driven cavity flow.

Five representative examples, including plane Poiseuille flow, 2D lid-driven cavity flow, polar cavity flow, flow past a stationary circular cylinder, and 3D lid-driven cavity flow, are presented for the comprehensive evaluation of the robustness and flexibility of HSLBM in simulating various incompressible viscous problems. It is shown in these examples that HSLBM can give more accurate results than SHSLBM for the same mesh size. When achieving comparable results, HSLBM consumes less virtual memory and computational time than SHSLBM. Moreover, the nice numerical stability of SHSLBM is well maintained in HSLBM, which makes it very promising in engineering problems with high Reynolds numbers.

APPENDIX: FORMULATIONS OF THREE-DIMENSIONAL HIGHLY ACCURATE SIMPLIFIED LATTICE BOLTZMANN METHOD

Three-dimensional Lagrange interpolating formulation reads

$$g(x,y,z) = \sum_{i=1}^{M}\sum_{j=1}^{N}\sum_{k=1}^{P} g\left(x_i,y_j,z_k\right)a_i(x)b_j(y)c_k(z), \tag{A1}$$

where the Lagrange basis polynomials a_i, b_j, and c_k are given by

$$a_i = \prod_{l=1,\, l\neq i}^{M} \frac{x-x_l}{x_i-x_l}, \tag{A2}$$

$$b_j = \prod_{l=1,\, l\neq j}^{N} \frac{y-y_l}{y_j-y_l}, \tag{A3}$$

$$c_k = \prod_{l=1,\, l\neq k}^{P} \frac{z-z_l}{z_k-z_l}. \tag{A4}$$

The three-dimensional predictor-corrector scheme is the same as those formulations in two-dimensional condition, i.e., Eqs. (16)–(19). The interpolation formulation in the predictor step [Eq. (20)] is changed into

$$f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t) = \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} f_\alpha^{eq}\left(\mathbf{r}_{ijk}, t-\delta_t\right)a_i(x-e_{\alpha x}\delta_t) \times b_j\left(y-e_{\alpha y}\delta_t\right)c_k(z-e_{\alpha z}\delta_t), \tag{A5}$$

where $\mathbf{r} = (x, y, z)$, $\mathbf{e}_\alpha = (e_{\alpha x}, e_{\alpha y}, e_{\alpha z})$; i, j, and k are the index of neighboring mesh points along x-, y-, and z-directions, respectively.

Following a similar interpolation procedure, the non-equilibrium distribution functions at locations required in the corrector step can also be calculated from three-dimensional Lagrange interpolation formulation.

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